

Module 15 - chapter 13

Project Deploy Nexus - Part 1

The Nexus installation shell commands can be found here: <https://gitlab.com/twn-devops-bootcamp/latest/15-ansible/ansible-projects/-/blob/main/nexus.sh>

If you are using version Nexus 3.70.1 + you must use Java version 17. Use the following code in the deploy-nexus.yaml file:

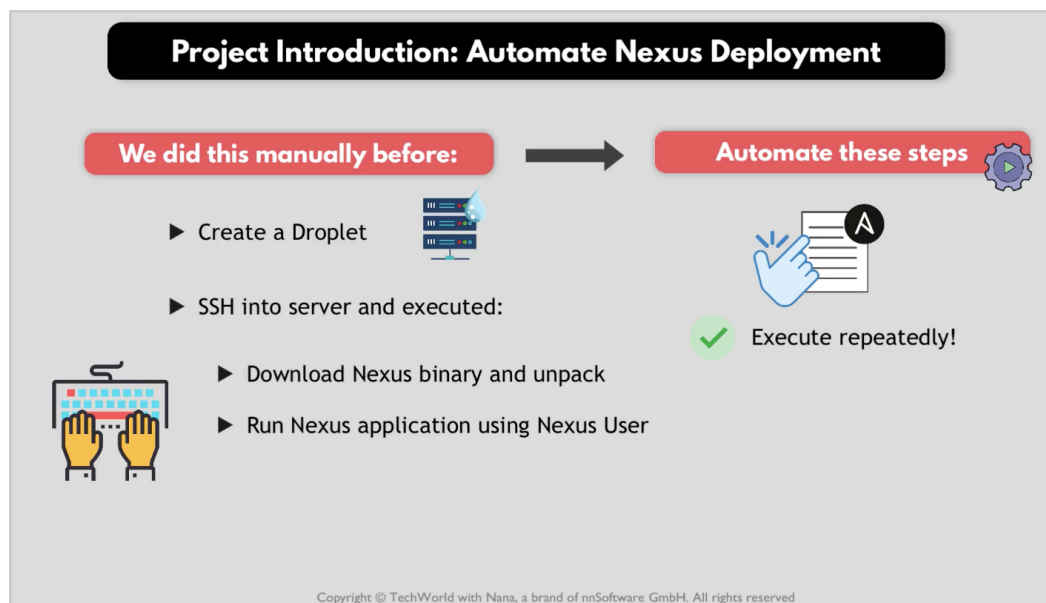
tasks:

- name: Install Java 17

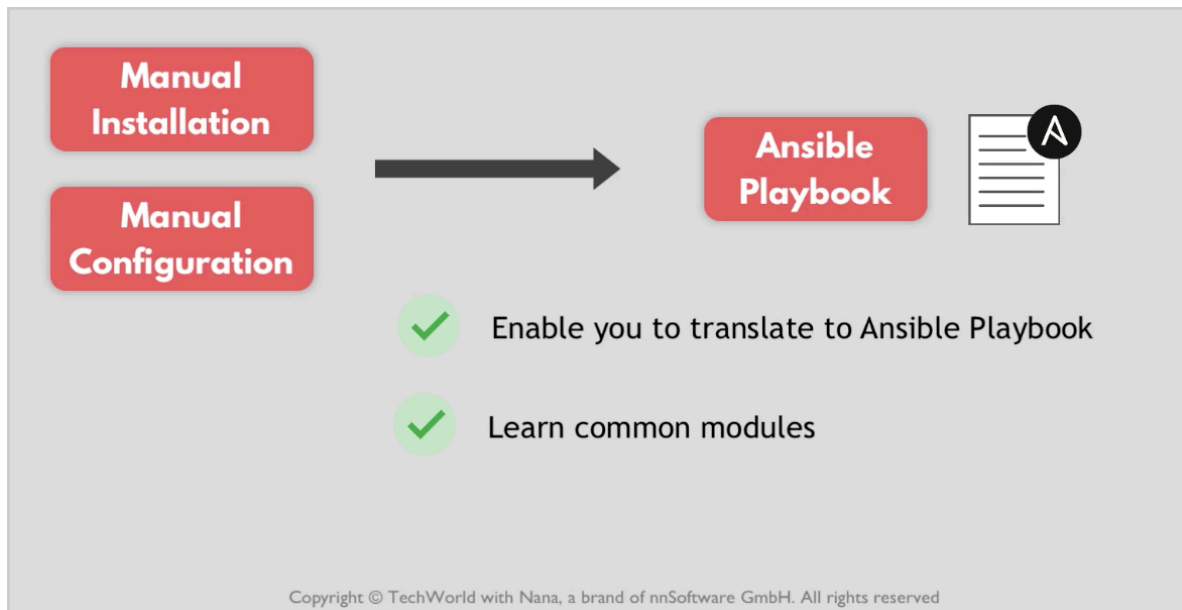
apt: name=openjdk-17-jre-headless

Project intro

We will automatise deploying nexus in a remote server



Manual installation of nexus was done in Module 6 - chapter 2



At the end of this chapter I should be able to take any installation guide and turn it into Ansible ❤️

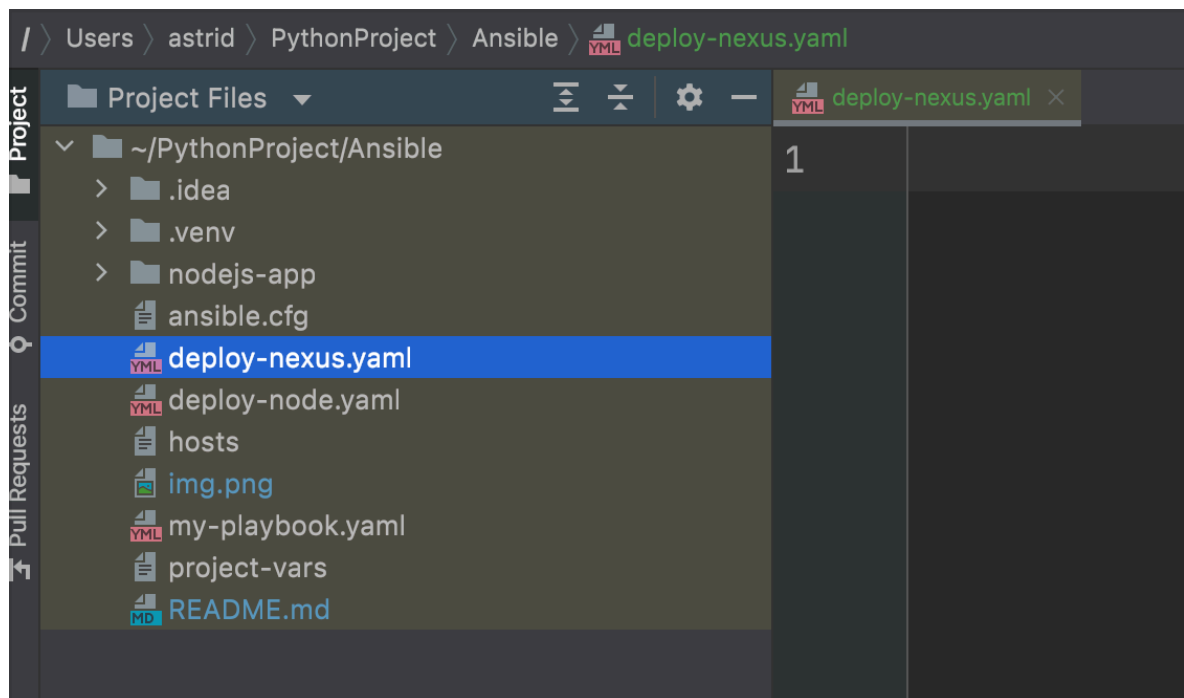
Preparation

Create new droplet

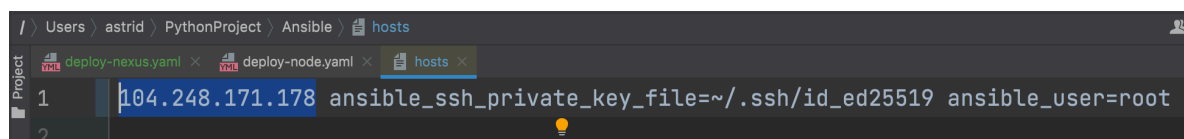
I named it -> nexus

Droplets				
Autoscale Pools				
🔍 Search by droplet name				
Name	IP Address	Created	Tags	
 nexus 1 GB / 25 GB / LON1 - Ubuntu 24.04 (LTS) x86	104.248.171.178 🌐	25 seconds ago	—	  ...
 jenkins-server 4 GB / 80 GB / LON1 - Ubuntu 24.04 (LTS) x86	165.22.116.21 🌐	8 months ago	—	  ...

Create a new playbook called 'deploy-nexus'

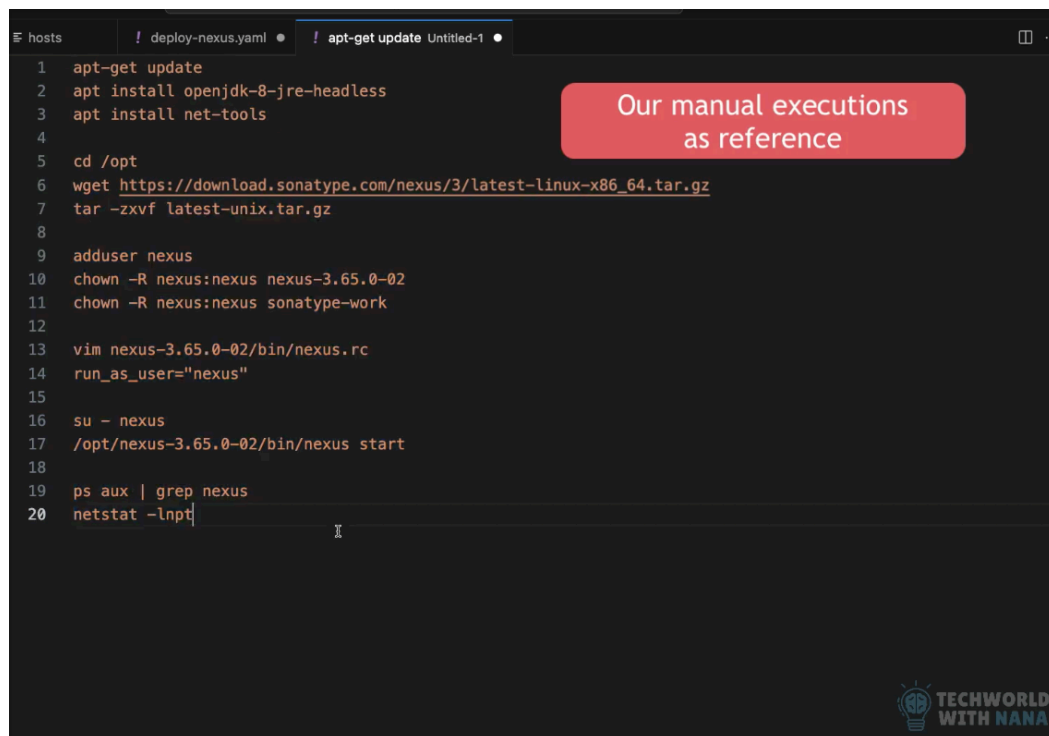


And we edit the hosts file with the IP address of the new droplet



Now let's write the playbook.

We want this playbook to be a guide of taking any linux commands or any steps of installing and deploying an application and translating that into a playbook, and because of that Nana created another file called 'apt-get update' where all the manual executions are so we can use it as a reference to install nexus

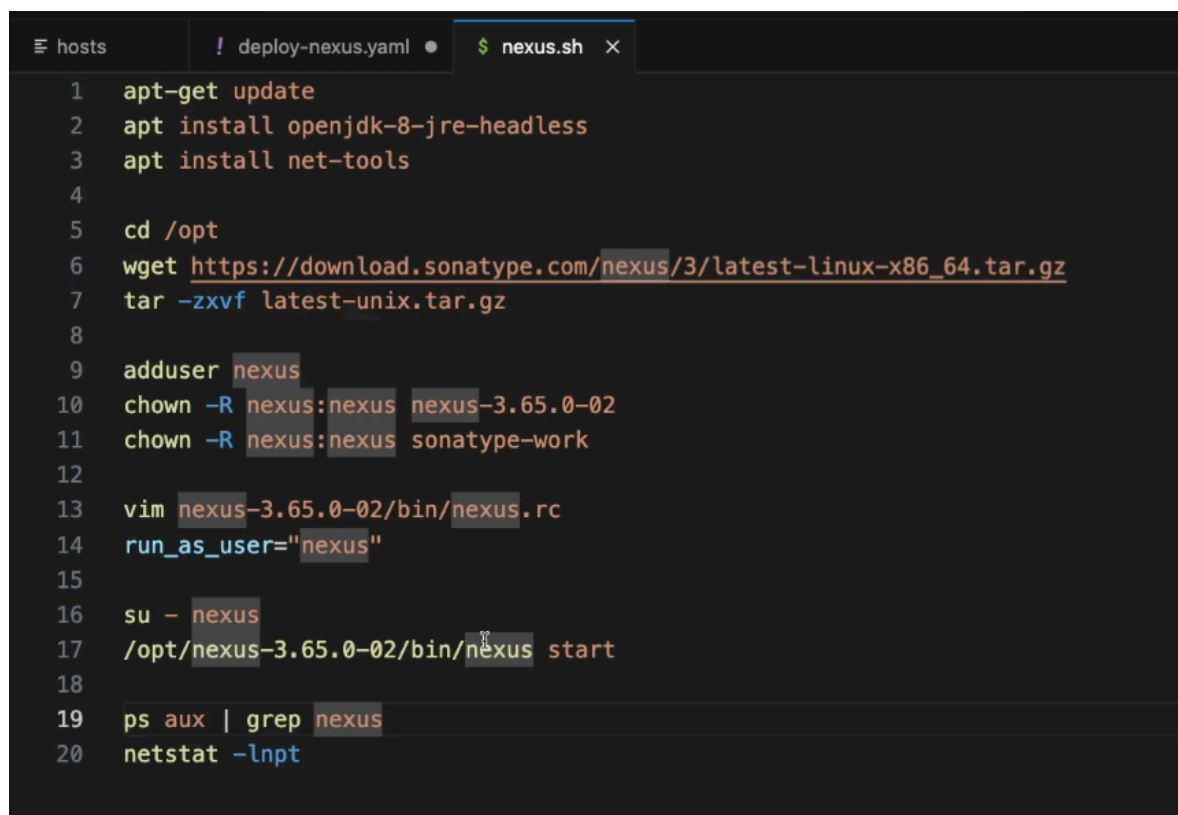


```
1 apt-get update
2 apt install openjdk-8-jre-headless
3 apt install net-tools
4
5 cd /opt
6 wget https://download.sonatype.com/nexus/3/latest-linux-x86_64.tar.gz
7 tar -zxvf latest-unix.tar.gz
8
9 adduser nexus
10 chown -R nexus:nexus nexus-3.65.0-02
11 chown -R nexus:nexus sonatype-work
12
13 vim nexus-3.65.0-02/bin/nexus.rc
14 run_as_user="nexus"
15
16 su - nexus
17 /opt/nexus-3.65.0-02/bin/nexus start
18
19 ps aux | grep nexus
20 netstat -lnpt
```

Our manual executions as reference

TECHWORLD WITH NANA

And then she rename it to 'nexus.sh'



```
1 apt-get update
2 apt install openjdk-8-jre-headless
3 apt install net-tools
4
5 cd /opt
6 wget https://download.sonatype.com/nexus/3/latest-linux-x86_64.tar.gz
7 tar -zxvf latest-unix.tar.gz
8
9 adduser nexus
10 chown -R nexus:nexus nexus-3.65.0-02
11 chown -R nexus:nexus sonatype-work
12
13 vim nexus-3.65.0-02/bin/nexus.rc
14 run_as_user="nexus"
15
16 su - nexus
17 /opt/nexus-3.65.0-02/bin/nexus start
18
19 ps aux | grep nexus
20 netstat -lnpt
```

This is my version using java 17 and nexus 3.84.1

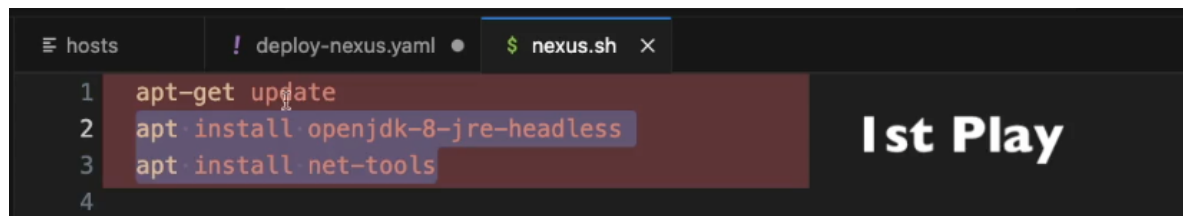
```

1 ▶ apt-get update
2 apt install openjdk-17-jre-headless
3 apt install net-tools
4
5 cd /opt
6 wget https://download.sonatype.com/nexus/3/nexus-3.84.1-01-linux-x86_64.tar.gz
7 tar -zxvf nexus-3.84.1-01-linux-x86_64.tar.gz
8
9 adduser nexus
10 chown -R nexus:nexus nexus-3.84.1-01
11 chown -R nexus:nexus sonatype-work
12
13 vim nexus-3.84.1-01/bin/nexus.rc
14 run_as_user="nexus"
15
16 $ - nexus
17 /opt/nexus-3.84.1-01/bin/nexus start
18
19 ps aux | grep nexus
20 netstat -lnpt

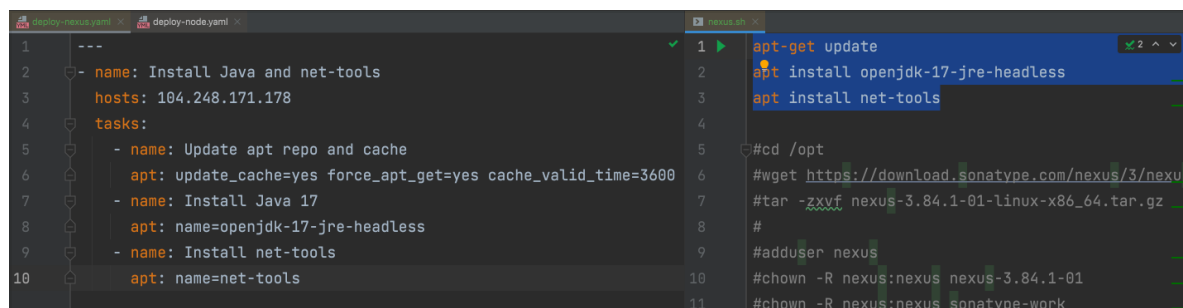
```

So first play is to install 2 packages:

- Java idk
- Net-tools



And here is the first play:



Let's write the second play

```
4
5 cd /opt
6 wget https://download.sonatype.com/nexus/3/latest-linux-x86_64.tar.gz
7 tar -zxvf latest-unix.tar.gz
8
```

2nd Play

Here we will change dir in the droplet and then download nexus in tar format (wget) and then we are unpacking it.

First task is to download nexus via https call. Download from internet.

```
1 ---
2 - name: Install Java and net-tools
3   hosts: 104.248.171.178
4   tasks:
5     - name: Update apt repo and cache
6       apt: update_cache=yes force_apt_get=yes
7     - name: Install Java 17
8       apt: name=openjdk-17-jre-headless
9     - name: Install net-tools
10      apt: name=net-tools
11
12 - name: Download and unpack Nexus installer
13   hosts: 104.248.171.178
14   tasks:
15     - name: Download Nexus
16
17
```

```
1 #apt-get update
2 #apt install openjdk-17-jre-headless
3 #apt install net-tools
4
5 cd /opt
6 wget https://download.sonatype.com/nexus/3/nexus-3.84.1-01-linux-x86_64.tar.gz
7 tar -zxvf nexus-3.84.1-01-linux-x86_64.tar.gz
8
9 #adduser nexus
10 #chown -R nexus:nexus nexus-3.84.1-01
11 #chown -R nexus:nexus sonatype-work
12 #
13 #vim nexus-3.84.1-01/bin/nexus.rc
14 #run_as_user="nexus"
15 #
16 #su - nexus
```

To download files from internet we have the module 'get_url'

"get_url" module

- ▶ Downloads files from HTTP, HTTPS or FTP to the remote server
- ▶ Separate module for Windows targets: "win_get_url"

And the module takes an URL as a parameter:

```
1  ---
2  - name: Install Java and net-tools
3    hosts: 104.248.171.178
4    tasks:
5      - name: Update apt repo and cache
6        apt: update_cache=yes force_apt_get=yes cache_val=
7      - name: Install Java 17
8        apt: name=openjdk-17-jre-headless
9      - name: Install net-tools
10       apt: name=net-tools
11
12 - name: Download and unpack Nexus installer
13   hosts: 104.248.171.178
14   tasks:
15     - name: Download Nexus
16       get_url:
17         url: https://download.sonatype.com/nexus/3/nexus-3.84.1-01-linux-x86_64.tar.gz
18         dest:
19
1  1 #apt-get update
2  2 #apt install openjdk-17-jre-headless
3  3 #apt install net-tools
4  4
5  5 cd /opt
6  6 wget https://download.sonatype.com/nexus/3/nexus-3.84.1-01-linux-x86_64.tar.gz
7  7 tar -zxvf nexus-3.84.1-01-linux-x86_64.tar.gz
8  8
9  9 #adduser nexus
10 10 #chown -R nexus:nexus nexus-3.84.1-01
11 11 #chown -R nexus:nexus sonatype-work
12 12 #
13 13 #vim nexus-3.84.1-01/bin/nexus.rc
14 14 #run_as_user="nexus"
15 15 #
16 16 #su - nexus
17 17 #/opt/nexus-3.84.1-01/bin/nexus start
18 18 #
19 19 #ps aux | grep nexus
```

And we can specify the destination as well, manually we cd into /opt/ folder (see line 5) so that is our destination in the play:

```
1  ---
2  - name: Install Java and net-tools
3    hosts: 104.248.171.178
4    tasks:
5      - name: Update apt repo and cache
6        apt: update_cache=yes force_apt_get=yes cache_val=
7      - name: Install Java 17
8        apt: name=openjdk-17-jre-headless
9      - name: Install net-tools
10       apt: name=net-tools
11
12 - name: Download and unpack Nexus installer
13   hosts: 104.248.171.178
14   tasks:
15     - name: Download Nexus
16       get_url:
17         url: https://download.sonatype.com/nexus/3/nexus-3.84.1-01-linux-x86_64.tar.gz
18         dest: /opt/
19
1  1 #apt-get update
2  2 #apt install openjdk-17-jre-headless
3  3 #apt install net-tools
4  4
5  5 cd /opt
6  6 wget https://download.sonatype.com/nexus/3/nexus-3.84.1-01-linux-x86_64.tar.gz
7  7 tar -zxvf nexus-3.84.1-01-linux-x86_64.tar.gz
8  8
9  9 #adduser nexus
10 10 #chown -R nexus:nexus nexus-3.84.1-01
11 11 #chown -R nexus:nexus sonatype-work
12 12 #
13 13 #vim nexus-3.84.1-01/bin/nexus.rc
14 14 #run_as_user="nexus"
15 15 #
16 16 #su - nexus
17 17 #/opt/nexus-3.84.1-01/bin/nexus start
18 18 #
19 19 #ps aux | grep nexus
```

And check how things are so far
-> ansible-playbook -i hosts deploy-nexus.yaml

```

TASK [Update apt repo and cache] *****
ok: [104.248.171.178]

TASK [Install Java 17] *****
changed: [104.248.171.178]

TASK [Install net-tools] *****
changed: [104.248.171.178]

PLAY [Download and unpack Nexus installer] *****

TASK [Gathering Facts] *****
ok: [104.248.171.178]

TASK [Download Nexus] *****
changed: [104.248.171.178]

PLAY RECAP *****
104.248.171.178 : ok=6  changed=3  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0

```

And check the droplet

```

Last login: Thu Aug  6 10:48:57 2026 from 140.228.47.252
root@nexus:~# ls /opt/
digitalocean nexus-3.84.1-01-linux-x86_64.tar.gz
root@nexus:~#

```

And nexus tar file is in the opt/ folder of the droplet.

We also have java installed

```

root@nexus:~# java
Usage: java [options] <main class> [<args>]
       (to execute a class)

```

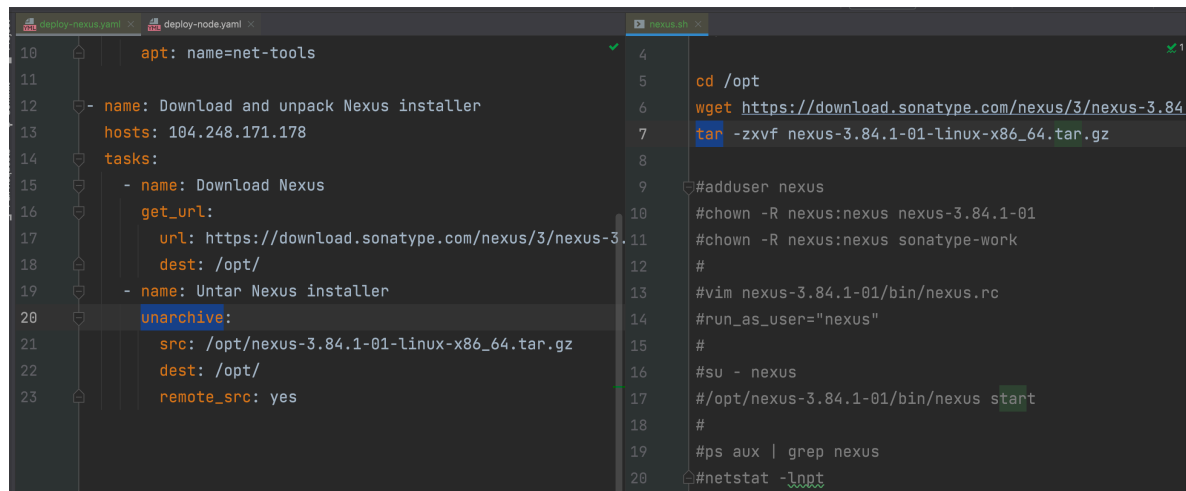
And netstat command

```

root@nexus:~# netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 nexus:ssh               198.211.111.194:44668   ESTABLISHED
tcp        0      0 nexus:ssh               198.211.111.194:44668   ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node  Path
unix   3      [ ]         STREAM     CONNECTED    6567
unix   3      [ ]         STREAM     CONNECTED    7607     /run/dbus/system_bus_socket
unix   2      [ ]         DGRAM      CONNECTED    2452
unix   3      [ ]         STREAM     CONNECTED    7611     /run/systemd/journal/stdout
unix   2      [ ]         DGRAM      CONNECTED    6573
unix   3      [ ]         STREAM     CONNECTED    7763     /run/systemd/journal/stdout
unix   3      [ ]         STREAM     CONNECTED    7608
unix   3      [ ]         DGRAM      CONNECTED    6578
unix   2      [ ]         DGRAM      CONNECTED    22383    /run/user/0/systemd/notify
unix   3      [ ]         STREAM     CONNECTED    7604
unix   2      [ ]         DGRAM      CONNECTED    7618
unix   3      [ ]         DGRAM      CONNECTED    6577
unix   3      [ ]         STREAM     CONNECTED    6568     /run/systemd/journal/stdout
unix   2      [ ]         DGRAM      CONNECTED    2260

```


Next task is to untar or unpack nexus.



The screenshot shows an IDE with two files open: `deploy-nexus.yaml` and `deploy-node.yaml`. The `deploy-nexus.yaml` file contains an Ansible playbook with two tasks: 'Download and unpack Nexus installer' and 'Untar Nexus installer'. The 'Untar Nexus installer' task uses the `unarchive` module with `src` and `dest` parameters, and `remote_src: yes`. To the right, a terminal window titled 'nexus.sh' shows the execution of commands: `cd /opt`, `wget https://download.sonatype.com/nexus/3/nexus-3.84.1-01-linux-x86_64.tar.gz`, `tar -zxvf nexus-3.84.1-01-linux-x86_64.tar.gz`, `#adduser nexus`, `#chown -R nexus:nexus nexus-3.84.1-01`, `#chown -R nexus:nexus sonatype-work`, `#`, `#vim nexus-3.84.1-01/bin/nexus.rc`, `#run_as_user="nexus"`, `#`, `#su - nexus`, `#/opt/nexus-3.84.1-01/bin/nexus start`, `#`, `#ps aux | grep nexus`, and `#netstat -lntp`.

And I use `remote_src` to tell ansible that the src is the droplet not my local machine.

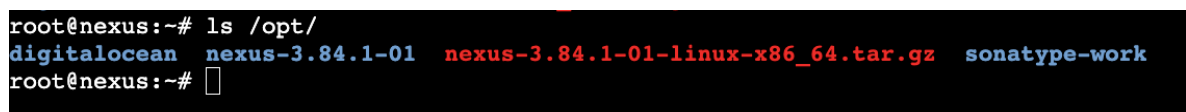
Let's run again the playbook

-> `ansible-playbook -i hosts deploy-nexus.yaml`



The screenshot shows the output of an Ansible playbook run. It displays two tasks: 'TASK [Download Nexus]' and 'TASK [Untar Nexus installer]'. The 'Download Nexus' task is marked as 'ok: [104.248.171.178]'. The 'Untar Nexus installer' task is marked as 'changed: [104.248.171.178]'. Below the tasks, a 'PLAY RECAP' section shows the overall status: '104.248.171.178 : ok=7 changed=2 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0'.

And the droplet has nexus unpacked



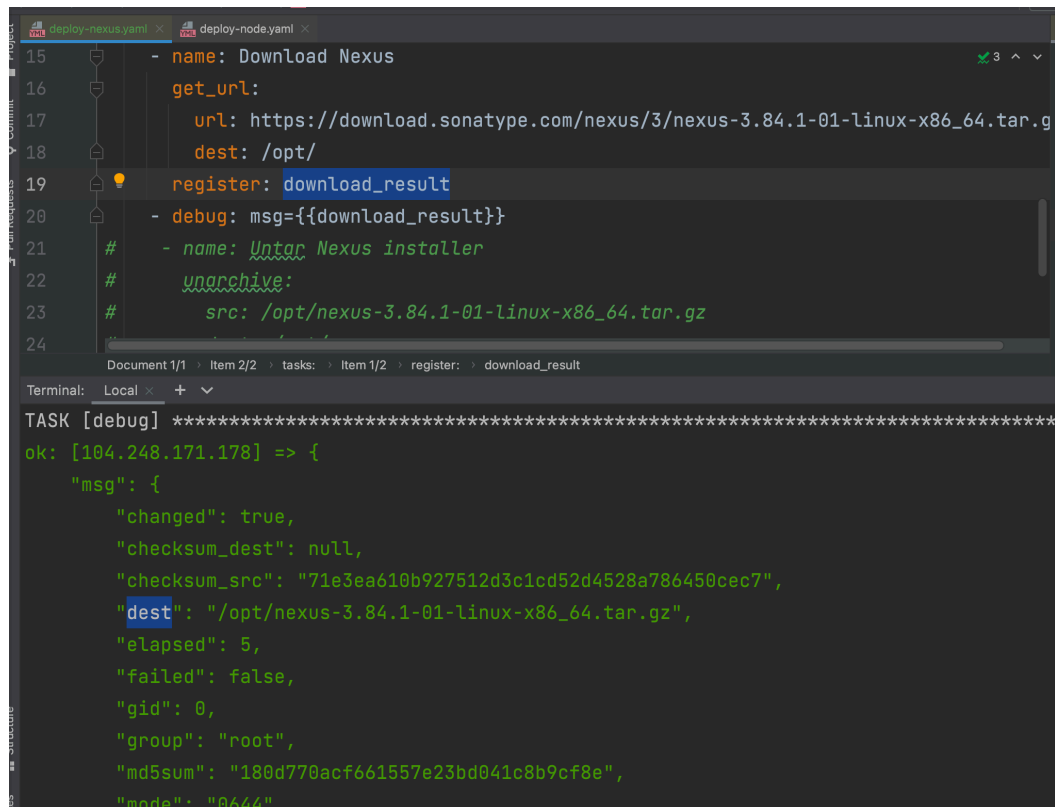
The screenshot shows a terminal window with the command `ls /opt/` executed. The output shows the following files and directories: `digitalocean`, `nexus-3.84.1-01`, `nexus-3.84.1-01-linux-x86_64.tar.gz`, and `sonatype-work`.

We have the nexus binaries -> `nexus-3.84.1-01`

And nexus storage like logs -> `sonatype-work`

We want to automate the installation, so we don't want to hardcode the version of nexus (Nana use latest but this is deprecated, we have to go to nexus download page and figure out the version ourselves <https://help.sonatype.com/en/download.html>)

So we can look at the return values of the 'get_url' task



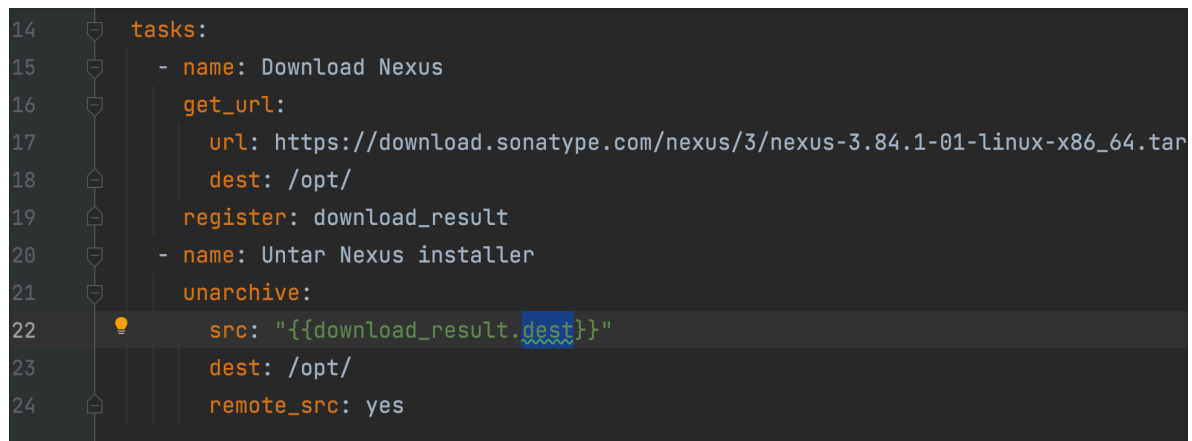
```
15 - name: Download Nexus
16   get_url:
17     url: https://download.sonatype.com/nexus/3/nexus-3.84.1-01-linux-x86_64.tar.g
18     dest: /opt/
19   register: download_result
20   - debug: msg={{download_result}}
21   # - name: Untar Nexus installer
22   #   unarchive:
23   #     src: /opt/nexus-3.84.1-01-linux-x86_64.tar.gz
24
```

```
Document 1/1 | Item 2/2 | tasks: | Item 1/2 | register: | download_result
```

```
Terminal: Local x + v
TASK [debug] *****
ok: [104.248.171.178] => {
  "msg": {
    "changed": true,
    "checksum_dest": null,
    "checksum_src": "71e3ea610b927512d3c1cd52d4528a786450cec7",
    "dest": "/opt/nexus-3.84.1-01-linux-x86_64.tar.gz",
    "elapsed": 5,
    "failed": false,
    "gid": 0,
    "group": "root",
    "md5sum": "180d770acf661557e23bd041c8b9cf8e",
    "mode": "0644",
```

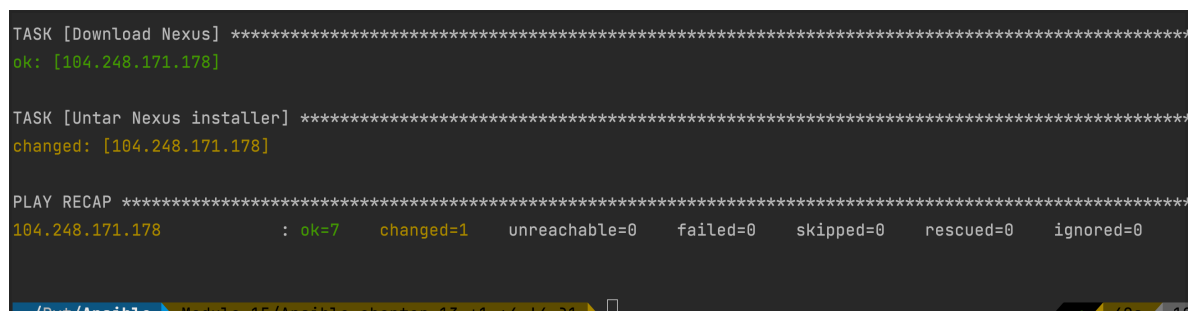
And we can see the path of the destination.

So we can use key 'dest' to get the path in the 'unarchive' task



```
14 tasks:
15   - name: Download Nexus
16     get_url:
17       url: https://download.sonatype.com/nexus/3/nexus-3.84.1-01-linux-x86_64.tar
18       dest: /opt/
19     register: download_result
20   - name: Untar Nexus installer
21     unarchive:
22       src: "{{download_result.dest}}"
23       dest: /opt/
24       remote_src: yes
```

And we run the playbook again:



```
TASK [Download Nexus] *****
ok: [104.248.171.178]

TASK [Untar Nexus installer] *****
changed: [104.248.171.178]

PLAY RECAP *****
104.248.171.178 : ok=7  changed=1  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
```

And check the droplet

```
root@nexus:~# ls /opt/
digitalocean nexus-3.84.1-01 nexus-3.84.1-01-linux-x86_64.tar.gz sonatype-work
root@nexus:~#
```

But we still have the version hardcoded when we download nexus, so let's create a variable there and set the value in project-vars:

```
15 - name: Download Nexus
16   get_url:
17     url: https://download.sonatype.com/nexus/3/nexus-{{nexus_version}}-linux-x86_64.tar.gz
18     dest: /opt/
19     register: download_result
20   name: Untar Nexus installer
21   unarchive:
22     src: "{{download_result.dest}}"
23     dest: /opt/
24     remote_src: yes

1 version: 1.0.0
2 location: /Users/astrid/Pyth
3 linux_name: astrid
4 user_home_dir: /home/{{linux
5
6 nexus_version: 3.84.1-01
```

```
12 - name: Download and unpack Nexus installer
13   hosts: 104.248.171.178
14   vars_files:
15     - project-vars
16   tasks:
17     - name: Download Nexus
18       get_url:
19         url: https://download.sonatype.com/nexus/3/nexus-{{nexus_version}}-linux-x86_64.tar.gz
20         dest: /opt/
21         register: download_result
```

And test everything works:
So I delete the file in the droplet

```
root@nexus:~# ls /opt/
digitalocean nexus-3.84.1-01 nexus-3.84.1-01-linux-x86_64.tar.gz sonatype-work
root@nexus:~# rm -rf /opt/nexus-3.84.1-01
root@nexus:~# rm -rf /opt/sonatype-work/
root@nexus:~# rm /opt/nexus-3.84.1-01-linux-x86_64.tar.gz
root@nexus:~# ls /opt/
digitalocean
```

And run the playbook again

```
TASK [Download Nexus] *****
changed: [104.248.171.178]

TASK [Untar Nexus installer] *****
changed: [104.248.171.178]

PLAY RECAP *****
104.248.171.178 : ok=7  changed=2  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
```

And check the droplet

```
root@nexus:~# ls /opt/  
digitalocean nexus-3.84.1-01 nexus-3.84.1-01-linux-x86_64.tar.gz sonatype-work  
root@nexus:~#
```

And so far so good.

"find" module

Nana wants to rename the nexus binary folder from nexus-3.84.1-01 to nexus

-> mv nexus-3.84.1-01 nexus

mv is Unix's single command for both — there's no separate "rename":

- Move (different directory, same name): mv file.txt /opt/
- Rename (same directory, different name): mv file.txt newname.txt
- Both at once: mv file.txt /opt/newname.txt

In her case she doesn't know the version of nexus that it goy installed (latest is now deprecated) so in her case she needs to find the folder she wants to rename but how if she doesn't know the version?

```
root@ubuntu-s-1vcpu-1gb-fra1-01:~# ls /opt/  
digitalocean nexus-3.65.0-02 nexus-3.65.0-02-unix.tar.gz sonatype-work  
root@ubuntu-s-1vcpu-1gb-fra1-01:~#
```



nexus-???

For cases like this ansible has a module 'find' using regex

"find" module

- ▶ Return a list of files based on specific criteria
- ▶ For Windows targets: "win_find" module

So let's get the name of the folder:

```
22     - name: Untar Nexus installer
23       unarchive:
24         src: "{{download_result.dest}}"
25         dest: /opt/
26         remote_src: yes
27     - name: Find nexus folder
28       find:
29         paths: /opt
30         pattern: "nexus-*"
31         file_type: directory
```

file_type by default is 'file' and we are looking for a folder so we need to specify that to 'directory'

And let's store the result and print it:

```
31     file_type: directory
32     register: find_result
33     - debug: msg={{find_result}}
```

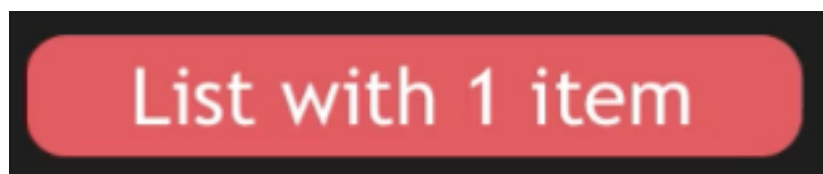
Document 1/1 > Item 2/2 > tasks: > Item 3/4 > register: > find_result

Terminal: Local < + v

TASK [debug] *****

```
ok: [104.248.171.178] => {
  "msg": {
    "changed": false,
    "examined": 4,
    "failed": false,
    "files": [
      {
        "atime": 1786016227.7827353,
        "blocks": 8,
        "ctime": 1786016230.7379653,
        "dev": 64769,
        "disk_usage_bytes": 4096,
        "gid": 0,
        "gr_name": "root",
        "inode": 524293,
        "isblk": false,
```

We get a list with one item, because we only have one directory with this name pattern



And in the path attribute we see the full path of the folder:

```
mode: 0755,
"mtime": 1786016230.7379653,
"nlink": 6,
"path": "/opt/nexus-3.84.1-01",
"pw_name": "root",
"rgrp": true,
```

Now we can use that key to get the folder and rename it:

```
25 - name: Find nexus folder
26   find:
27     paths: /opt
28     pattern: "nexus-*"
29     file_type: directory
30     register: find_result
31 - debug: msg={{find_result}}
32 - name: Rename nexus folder
33   shell: mv {{find_result.files[0]}}
```

Access list item with index

0 index for first item

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL zsh + -

```
"failed": false,
"files": [
  {
    "atime": 1707236377.497423,
    "ctime": 1707236377.737425,
    "dev": 64769,
    "gid": 0,
    "gr_name": "root",
```

```
32   register: find_result
33 - name: Rename nexus folder
34   shell: mv {{find_result.files[0].path}} /opt/nexus
```

Document 1/1 > Item 2/2 > tasks: > Item 4/4 > shell: > mv {{find_result.fil...

Terminal: Local x + v

```
"nlink": 6,
"path": "/opt/nexus-3.84.1-01",
"pw_name": "root",
"rgrp": true,
"roth": true,
```

The new name is 'nexus' and we need to specify the full path -> /opt/nexus

I rerun the playbook and checked the droplet

```
root@nexus:~# ls /opt/
digitalocean nexus nexus-3.84.1-01-linux-x86_64.tar.gz sonatype-work
root@nexus:~#
```

The folder has been renamed to 'nexus'

Conditionals in Ansible

If we rerun the playbook we will have an error when it reaches the 'renaming' of the folder

```

TASK [Rename nexus folder] *****
fatal: [134.122.82.225]: FAILED! => {"changed": true, "cmd": "mv /opt/nexus-3.65.0-02 /opt/nexus", "delta": "0:00:00.006324", "end": "2024-02-06 17:10:47.399071", "msg": "non-zero return code", "rc": 1, "start": "2024-02-06 17:10:47.392747", "stderr": "mv: cannot move '/opt/nexus-3.65.0-02' to '/opt/nexus/nexus-3.65.0-02': Directory not empty", "stdout_lines": ["mv: cannot move '/opt/nexus-3.65.0-02' to '/opt/nexus/nexus-3.65.0-02': Directory not empty"], "stdout": "", "stdout_lines": []}

PLAY RECAP *****
134.122.82.225      : ok=8    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0

```

because it will untar nexus and create again the file with the version

```

root@ubuntu-s-1vcpu-1gb-fra1-01:~# ls /opt/
digitalocean nexus nexus-3.65.0-02 nexus-3.65.0-02-unix.tar.gz sonatype-work

```

Then it will take that folder and it will try to rename it to nexus but nexus folder already exists and is not empty

```

root@ubuntu-s-1vcpu-1gb-fra1-01:~# ls /opt/nexus
NOTICE.txt OSS-LICENSE.txt PRO-LICENSE.txt bin deploy etc lib nexus-3.65.0-02 public replicator system
root@ubuntu-s-1vcpu-1gb-fra1-01:~#

```

So it fails.

If I run the playbook again, it doesn't untar again because it finds the two folders (nexus binary and nexus storage) but it fails at renaming for the reason already explained.

'shell' module does not check state. **!!** so it gets executed in every run.

So we need to check if the state we want exist to skip the task -> conditional

Conditionals

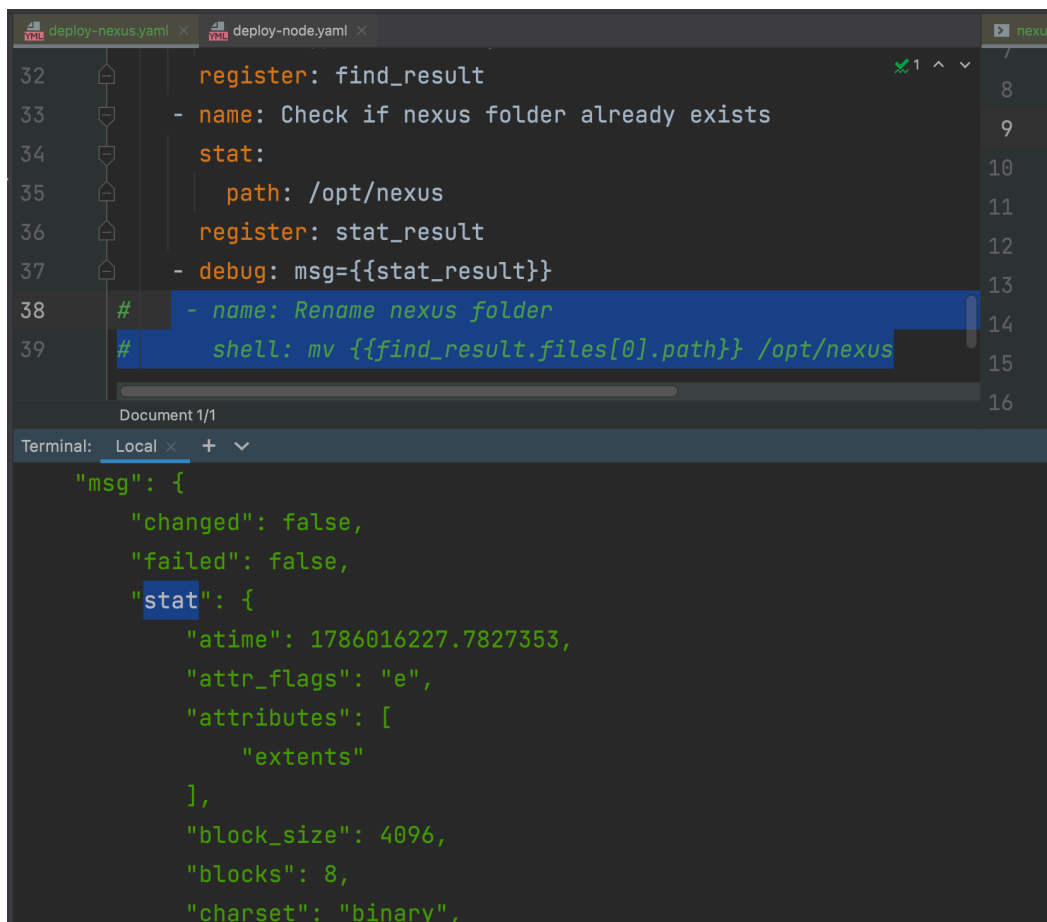
► Execute task depending on some condition

We want to check if a folder or file already exist so we use a module called 'stat' to get statistics of the folder or file

"stat" Module

► Retrieve file or file system status

Here is the code



The screenshot shows a code editor with two tabs: 'deploy-nexus.yml' and 'deploy-node.yml'. The 'deploy-nexus.yml' tab is active, showing lines 32 to 39 of an Ansible playbook. Line 32 sets 'register: find_result'. Line 33 is a task to check if the nexus folder exists. Line 34 sets 'stat: path: /opt/nexus'. Line 35 sets 'register: stat_result'. Line 36 is a debug task. Line 37 is a comment. Line 38 is a task to rename the nexus folder. Line 39 is a comment. The 'Terminal' panel at the bottom shows the output of the 'stat' module, which is a dictionary with keys 'changed', 'failed', 'stat', 'atime', 'attr_flags', 'attributes', 'block_size', 'blocks', and 'charset'.

```
32      register: find_result
33      - name: Check if nexus folder already exists
34        stat:
35          path: /opt/nexus
36        register: stat_result
37      - debug: msg={{stat_result}}
38      # - name: Rename nexus folder
39      #   shell: mv {{find_result.files[0].path}} /opt/nexus
```

```
"msg": {
  "changed": false,
  "failed": false,
  "stat": {
    "atime": 1786016227.7827353,
    "attr_flags": "e",
    "attributes": [
      "extents"
    ],
    "block_size": 4096,
    "blocks": 8,
    "charset": "binary",
```

And 'stat' module returns a 'stat' key value where the value is a dictionary. In the dictionary there is a key 'exists'

```

"stat": {
  "atime": 1786016227.7827353,
  "attr_flags": "e",
  "attributes": [
    "extents"
  ],
  "block_size": 4096,
  "blocks": 8,
  "charset": "binary",
  "ctime": 1786017892.9027839,
  "dev": 64769,
  "device_type": 0,
  "disk_usage_bytes": 4096,
  "executable": true,
  "exists": true,
  "gid": 0,

```

In this way we will know if the folder exists or not

```

31   - name: Check if nexus folder already exists
32     stat:
33       path: /opt/nexus
34       register: stat_result
35   - debug: msg={{stat_result.stat.exists}}
36   - name: Rename nexus folder
37     shell: mv {{find_result.files[0].path}} /opt/nexus

```

Conditionals

- ▶ Execute task depending on some condition
- ▶ Condition based on the value of a fact, a variable, or the result of a previous task

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```

TASK [debug] *****
ok: [134.122.82.225] => {
  "msg": {
    "changed": false,
    "failed": false,
    "stat": {
      "atime": 1707239326.4641423,
      "attr_flags": "e",
      "attributes": [
        "extents"
      ],
      "block_size": 4096,
      "blocks": 8,

```

And to define the condition to whether execute the shell command or not we

use 'when'

```
27 paths: /opt
28 pattern: "nexus-*"
29 file_type: directory
30 register: find_result
31 - name: Check nexus folder stats
32   stat:
33     path: /opt/nexus
34     register: stat_result
35 - debug: msg={{stat_result.stat.exists}}
36 - name: Rename nexus folder
37   shell: mv {{find_result.files[0].path}} /opt/nexus
38   when: not stat_result.stat.exists
```

"when"

- ▶ Applies to a single task
- ▶ No double curly braces needed here!
- ▶ The when statement applies the test

So when 'stat_result.stat.exists' is **not** 'true' then execute 'shell' if it is 'true' then skip the task.

```
33 - name: Check nexus folder stats
34   stat:
35     path: /opt/nexus
36     register: stat_result
37 - name: Rename nexus folder
38   shell: mv {{find_result.files[0].path}} /opt/nexus
39   when: not stat_result.stat.exists
```



Re-run the playbook

```
TASK [Check nexus folder stats] *****
ok: [104.248.171.178]

TASK [Rename nexus folder] *****
skipping: [104.248.171.178]

PLAY RECAP *****
104.248.171.178 : ok=9  changed=0  unreachable=0  failed=0  skipped=1  rescued=0  ignored=0
```

The task was skipped this time 🤔

So use 'when' conditionals for 'shell' and 'command' modules.

So at this point we have:

- Install java
- Install netstat
- Download nexus
- Unpacked nexus
- Rename nexus binary folder

